

Education

- 2011–2016 **PhD, Mechanical Engineering**, *University of Minnesota Twin Cities, USA*
Minor: Aerospace Engineering and Mechanics
Thesis: Bubbly flow physics for applications in next generation aerated hydro turbines and underwater transport (Supervisors: Prof. Roger E A Arndt, Prof. Jiarong Hong)
- 2011–2013 **MS, Mechanical Engineering**, *University of Minnesota Twin Cities, USA*
Minor: Aerospace Engineering and Mechanics
Specialization: Aerosol Sciences and Particle Technology, Aerosol Instrumentation
- 2007–2011 **B.Tech, Mechanical Engineering**, *Indian Institute of Technology Delhi (IITD), New Delhi*

Awards & Fellowships

Awarded by Government of India Agencies

- 2021 National level Winner of **Toycathon-2021** under the Digital Toycathon category. Awarded by *All India Council for Technical Education (AICTE)* in colLaboratory development with several ministries.
- 2020 **COVID Innovation Challenge**, *National Innovation Foundation (NIF)*, Department of Science and Technology, Government of India.
- 2018 **Early Career Research Award**, *Science and Engineering Research Board (SERB)*, Department of Science and Technology, Government of India.
- 2005-2011 **National Talent Search (NTS) Scholarship** awarded by *National Council for Educational Research and Training (NCERT)*, New Delhi.

Awarded by Other Entities/Organizations

- 2025 **Best Oral Presentation Award** for the paper presented by my PhD student at the International Conference on Advances in Sustainable Solutions for Energy Transitions, organized by the *School of Energy Science and Engineering, IIT Guwahati*, January 2025
- 2021 Recipient of the **Nation Builder's Award** on Teacher's day by *Rotary International, Dehradun Rotary Club*.
- 2020 National level Winner of **COVID-19 Open Innovation Challenge**. Awarded by *Foundation for Integrated Support and Solution (FISS)*, Assam in collaboration with *Gujarat Technological University, India*.
- 2020 **Teaching Excellence Award**, by *University of Petroleum and Energy Studies (UPES)*, Uttarakhand, India.
- 2020 Winner of **Best Paper Award at Tech Symposium**. Awarded by *UPES, Dehradun* for the research on water quality enhancement in hydropower impoundments.
- 2019 **Research Excellence Award**, by *University of Petroleum and Energy Studies (UPES)*, Uttarakhand, India.
- 2019 **Teaching Excellence Award**, by *University of Petroleum and Energy Studies (UPES)*, Uttarakhand, India.
- 2011 Bagged third place and honorable mention at the **International Metallographic Contest**, by *American Society of Metals*. Based on undergraduate research.

Awarded at University of Minnesota, USA

- 2016 Recipient of the **Preparing Future Faculty Certificate** by the *Graduate School* for having undergone a rigorous year long course on teaching and pedagogy, along with a practicum.
- 2015 Recipient of the **Edward Silberman Award** for excellence in scientific research by *College of Science and Engineering* with a cash prize of 10,000 USD.
- 2015 Recipient of the **Doctoral Dissertation Fellowship** by the *Graduate School*. Awarded to selected best dissertations across the university with funding support for the final year of PhD.
- 2015 Recipient of the **Student Fund Award** by the *Saint Anthony Falls Laboratory (SAFL)* Awarded to selected graduate researchers for conference travel.
- 2014 Recipient of the **Frank and Julie Tsai Travel Award** by the *Saint Anthony Falls Laboratory (SAFL)* Awarded to selected graduate researchers for conference travel.
- 2011 Recipient of the **Mechanical Engineering Department Fellowship** during the initial year of the *Graduate School*.

Awarded at IIT Delhi

- 2011 Recipient of the **BOSS Award** at the convocation for best hardcore experimental project from *Department of Mechanical Engineering* alongwith with a cash prize.
- 2009 **Summer Undergraduate Research Award** by *Industrial Research and Development (IRD) Unit*.

Formal Training and Education in Indian Knowledge System

- 2026 **Advanced Studies in IKS**, *Bhandarkar Oriental Research Institute (Bharat Vidya)*
 - Indian Temple Architecture
 - Veda Vidya: Vedas to Vedāṅgas
 - कुट्टक-चक्रवालन्यासविधि — Solutions of Linear and Quadratic Indeterminate Equations
 - Ayurveda: The Science of a Healthy Life
- 2025–2026 **Advanced Studies in Indian Astronomical Texts**, *Vyoma Sanskrit Pathshala, Siddhanta Knowledge Foundation & Samskrit Promotion Foundation*
 - Vyoma Sanskrit Pathshala: A bird's eye view of Siddhānta Jyotiṣa (Ongoing)
 - Samskrit Promotion Foundation: Sūrya-siddhānta (Madhyamādhikārah), Āryabhaṭīyam Gaṇitapādaḥ, Golaparibhāṣā (Ongoing)
 - Siddhanta Knowledge Foundation: IKS perspective on Sustainability: Agriculture (Ongoing)
- 2025–2026 **Foundations of Indian Astronomy** (नक्षत्रविद्या), *Anaadi Foundation course on SWAYAM*
Completed a certification course in foundations of Indian Astronomy on SWAYAM portal.
- 2025 **Community Education Program**, *Hindu University of America, Orlando, USA*
Completed a set of five quarter-long courses:
 - History and Methods of Indian Astronomy (Autumn 2026)
 - Exploring Hinduism: Cosmology and Calendar (Winter 2025)
 - Ayurveda 101: Wisdom of Well-being (Winter 2025)
 - Introduction to Vedic Chanting (Summer 2025)
 - Exploring Chandas (Summer 2025)
- 2024–2025 **Basic Course on Panchāṅga**, *Madras Sanskrit College, Chennai*
Completed Basic Panchāṅga (Jyotiṣa) training covering the five aṅgas—vāra, tithi, nakṣatra, yoga, karaṇa - and practical time-period computations.

- 2023–2025 **Advanced Certificate Programme in Sanskrit Language**, *Central Sanskrit Univ., New Delhi*
Completed a set of three courses : द्वितीयदीक्षा-१, द्वितीयदीक्षा-२, श्रवणकौशलम्-२.
- 2023–2025 **Sanskrit Correspondence Program (पत्राचारद्वारा संस्कृतम्)**, *Sanskrit Bharati, New Delhi*
Completed a structured four-level program (each six months, with formal assessment):
- प्रवेशः — October 2023 (With Distinction)
 - परिचयः — March 2024 (With Distinction)
 - शिक्षा — September 2024 (With Distinction)
 - कोविदः — March 2025 (With Distinction)
- 2023 **Certificate in Sanskrit Language**, *Uttarakhand Open University, Haldwani*
Formal certification in Sanskrit language fundamentals through two courses on सहज संस्कृत बोध (CSL101, CSL102)
- 2023 **Online Courses in Sanskrit, Indian Mathematics, and Astronomy**, *NPTEL – SWAYAM*
- Introduction to Basic and Intermediate Spoken Sanskrit, IIT Kharagpur (Completed with Distinction)
 - Mathematics in India: From Vedic Period to Modern Times, IIT Bombay
 - Astronomy in Ancient, Medieval, and Early Telescopic Era of India, IIT Kanpur
- 2005 **Certificate of Merit in Sanskrit**, *Central Board of Secondary Education (CBSE)*
Awarded for outstanding academic performance in Sanskrit at the secondary level in AISSE-05. Certificate of honour and gold medal was awarded by Ramakrishna Mission Vidyapith, Deoghar, Jharkhand.

Work Experience

Indian Institute of Technology Bombay (IITB) Mumbai, India

June 2026 – **Associate Professor**, *Centre for Traditional Indian Knowledge & Skills*, IIT Bombay, Powai, Mumbai, India.

University of Petroleum and Energy Studies (UPES) Dehradun, India

March 2025 – **Professor**, *Mechanical & Aerospace Engineering, School of Engineering*, Energy Acres, Bidholi, Dehradun.

March 2023 – **Sr. Associate Professor**, *Mechanical & Aerospace Engineering, School of Engineering*, Energy Acres, Bidholi, Dehradun.

Dec 2021 – **Head of the Department**, *Mechanical & Aerospace Engineering, School of Engineering*, Energy Acres, Bidholi, Dehradun.

July 2021 – **Associate Professor**, *Mechanical & Aerospace Engineering, School of Engineering*, Energy Acres, Bidholi, Dehradun.

July 2017 – **Assistant Professor (Selection Grade)**, *Mechanical Engineering, School of Engineering*, Energy Acres, Bidholi, Dehradun.

College of Engineering Roorkee, Uttarakhand, India

June 2016 – **Assistant Professor**, *Department of Mechanical Engineering*

University of Minnesota Twin Cities, USA

Aug 2011 – **Graduate Research Assistant, Saint Anthony Falls Laboratory**

Jan 2016 Led two distinct U.S. government-funded projects at Cavitation and Bubbly Flows research group in the field of fluid mechanics, focusing on the following applications:

Naval Defense: The Dynamics and Control of Supercavitating Vehicles, funded by the Office of Naval Research, United States (Grant # N000140910141).

Hydro Turbines: Turbine aeration design software for mitigating adverse environmental impacts resulting from conventional hydropower turbines, funded by the U.S. Department of Energy (Grant # DOE-UMN-0005416).

Teaching Experience

IKS Courses Developed IKS Foundations of Biomimicry and Nature-inspired Engineering (Aug-Dec 2025), Indian Knowledge System in Science and Engineering (Jan-May 2023; Aug-Dec 2023), Foundations of Āyurveda (Jan-May 2026; Ongoing)

Courses Taught Foundations of Traditional Indian Knowledge and Skills (Ongoing), Fluid Mechanics, Applied Fluid Mechanics (Automotive Design), Heat Transfer, Heat Transfer for Aerospace Applications, Turbomachines, Fluid Mechanics Laboratory, Thermal Laboratory, Fluid Machines Laboratory, Thermodynamics, Applied Thermodynamics, Aerodynamics, Advanced Fluid Dynamics, Advanced Fluid Dynamics and Heat Transfer

Others Programming for Engineers (MATLAB), Heat Transfer for Aerospace Applications

Laboratory Development Experience

MultiPhase Flows Laboratory Established the MultiPhase Flows Laboratory at UPES, equipped with state-of-the-art instrumentation for advanced flow measurement and analysis. The facility features high-speed imaging cameras with precision optics, high-power lasers for flow illumination, and sophisticated velocity measurement devices. It includes hydraulic flumes, a small-scale water tunnel, and experimental tanks designed for fluid dynamics research. Additionally, the laboratory houses air injection and flow measurement systems, a rotating disk apparatus for turbine runner testing, and a customizable screw turbine setup for laboratory-scale evaluations. To support rapid innovation and prototyping, the facility is further equipped with advanced manufacturing tools for fabricating turbine components.

Outdoor High-Velocity Water Flow Facility The outdoor research laboratory was established in the hilly region of Maldeota, Dehradun, through a grant from Uttarakhand Jal Sansthan under a Memorandum of Understanding (MoU) for research and development. The facility focuses on pioneering research in hydrokinetic turbines and the development of sustainable water-lifting technologies tailored for hilly terrains.

Bajaj Engineering Skill Training Centre As the Head of the Department of Mechanical and Aerospace Engineering, led a strategic initiative to secure external grants exceeding 20 crores from Bajaj Auto Ltd. Currently engaged in the establishment and advancement of state-of-the-art laboratories in Mechatronics, Motion Control & Sensor Technology, Automation & Robotics, and Industry 4.0 & Smart Manufacturing.

Funded Research Grants

Grants Funded by Government Agencies

2018 Ventilation augmented supercavitating technology for next-generation hydropower, **Science and Engineering Research Board (SERB), Department of Science and Technology**, (As Sole PI), Funding amount: Rs. 40.3 lakhs

2020 Design, Fabrication and Pilot Study of a Comfortably Vented Indigenously Designed (COVID) Fabric Helmet to Curb Infection Spread in Community Settings such as Education and Healthcare, **National Innovation Foundation (NIF), Department of Science and Technology**, (As PI), Funding amount: Rs. 1.5 lakhs

- 2022 Early Science Technology Engineering Education & Mathematics of India Research Centre,, **All India Council for Technical Education (AICTE), Ministry of Education, Government of India**, (As PI), Funding amount: Rs. 18.5 lakhs.
- 2022 Creation of IKS-based Hands-On Activities/Indian Toys for High School Science Education and NCERT-based Translated Materials in Odia for High School Science Education., **All India Council for Technical Education (AICTE), Ministry of Education, Government of India**, (As co-Project Investigator)
- 2024 IKS Wiki Articles in Mathematics, Astronomy, and Physical Sciences. Research Internship Grant, 2024., **All India Council for Technical Education (AICTE), Ministry of Education, Government of India**, (As Project Investigator)
- 2025 Financial Grant for organizing “National Space Convention 4.0 / International Space Convention 1.0”., **Indian Space Research Organisation (ISRO), Department of Space, Government of India**, (As Chairperson & Principal Organizer), Funding amount: Rs. 1 Lakh

Internal Seed Grants funded by UPES

- 2018 In-situ monitoring of industrial bubbly flows and bubble size control via continuous feedback approach, **UPES SEED Research Grant**, (As co-PI), Funding Amount: Rs. 3.15 Lakhs.
- 2018 Novel Strategies to induce bubble coalescence for industrial and naval applications, **UPES SEED Research Grant**, (As PI), Funding Amount: Rs. 2.4 Lakhs.
- 2021 Experimental Investigation of ventilated supercavitation at high cavitator temperature, **UPES SEED Research Grant**, (As co-PI), Funding Amount: Rs. 3.75 Lakhs.
- 2021 Novel technology of water lifting technique and efficiency enhancement of hydro turbines in remote hilly areas, **UPES SEED Research Grant**, (As co-PI), Funding Amount: Rs. 2.75 Lakhs.

Current Research Interests

Indian Knowledge	Nature-inspired Engineering in Indian Texts, Biomimicry and Bio-inspiration and its Indian roots, Principles of Mechanics in Indian Texts and Tradition, Āyurveda, Indian Mathematics and Astronomy
Engg. Education	Active Learning in Engineering, Innovative pedagogy in engineering classrooms, Biomimetics in engineering curriculum.

Journal Publications

[Total: 57 ; SCI/SCIE : 39 ; ESCI : 7 ; SCOPUS : 10; Others (GOI Policy Journals): 1]

- 2026 Dhiman, P., Chaudhary, A., Rautela, N., Karn, A. (2026). *Enhancing Hydropower Sustainability through Air Injection: Experimental Reduction of Sediment Erosion in Francis Turbine Guide Vanes* **Renewable Energy and Sustainable Development**, **12**(1), 248-267. (Scopus)
- 2026 Kumar, A., Karn, A., McGregor, C., Singh, V. P. (2026). *Experimental Thermal and Exergy Performance of a Flat Spiral External Receiver for a Scheffler Dish Concentrator* **Energy**, 141457. DOI: 10.1016/j.energy.2026.141457. IF: 9.4 (Q1)
- 2026 Uniyal, V., Singh, V. P., Singh, S., Karn, A. (2026). *Performance Characterisation of an Archimedes Screw Turbine Using Experimental Data and Machine Learning Regression Techniques* **Engineering Research Express**, **8**, 035532. IF: 1.6 (ESCI)
- 2026 Kumar, A., Karn, A., McGregor, C., Singh, V. P. (2026). *Life Cycle Sustainability Assessment of Box Solar Cooking Technologies: Comprehensive Comparative Analysis with Traditional Fuels*. **Renewable and Sustainable Energy Reviews**, **228**, 116581 . IF: 16.3 (SCI)
- 2025 Gaurav, K., Venkatesh, N., Raj, A., Karn, A. (2025). *Supercavity Enhancement in High-Speed Heavyweight Torpedoes: Exploring the Role of Secondary Cavitator*. **Journal of Marine Science and Application**. IF: 2.1 (ESCI)

- 2025 Donga, R. K., Karn, A. (2025). *Improving the thermal performance of parabolic trough solar collectors by incorporating cylindrical attachments within the absorber tube.* **Applied Thermal Engineering**, **273**, 126587. IF: 6.9 (SCI)
- 2025 Uniyal, V., Singh, V. P., Karn, A. (2025). *Comparative Life Cycle Assessment of Micro Water Turbines: Evaluating Potential for Remote Power Generation.* **Sustainable Energy Technologies and Assessments**, **73**, 104107. IF: 7.1 (SCI)
- 2025 Kumar, A., Donga, R. K., Karn, A. (2025). *Large eddy simulation of rectangular synthetic jets and insights into vortex bifurcation dynamics.* **Journal of Visualization**. IF: 2.3 (SCI)
- 2025 Dhiman, P., Bhat, A., Karn, A. (2025). *Multiphase flow simulations to explore novel technique of air injection to mitigate silt erosion in hydro turbines.* **Journal of Applied Fluid Mechanics**, **18(2)**, 468-484. IF: 1.3 (SCIE)
- 2025 Uniyal, V., Karn, A., Singh, V. P. (2025). *Parametric Optimization of Archimedes Screw Turbines for High-Efficiency and Sustainable Hydropower.* **Energy Sources Part A: Recovery, Utilization, and Environmental Effects**, **47(2)** IF: 2.2 (SCI).
- 2025 Karn, A., Verma, M., Singh, R. K., Mittal, G. (2025). *The Saga of the Dance School: Digital Story-telling in the Fluids classroom.* **Journal of Engineering Education Transformations**, **38(4)**, 75–87. CiteScore: 1.8 (Scopus)
- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V. P. (2025). *A critical analysis of paradigm shifts of individual solar cooking to institutional solar cooking technologies..* **Renewable and Sustainable Energy Reviews**, **218**, 115840. IF: 16.3 (SCI)
- 2024 Sanjeevaih, P. K., Karn, A. (2024). *Enhancement of dissolved oxygen level in the reservoir downstream using air injection techniques in hydroturbines.* **Energy Sources, Part A**, **46(1)**, 9934-9953. IF: 2.2 (SCI)
- 2024 Uniyal, V., Singh, V. P., Karn, A. (2024). *Parametric optimization of Archimedes screw turbine by response surface methodology and artificial neural networks.* **Renewable Energy and Sustainable Development**, **10(2)**, 306–318 (Scopus)
- 2024 Gaurav, K., Nagothu, V., Karn, A. (2024). *A comparative assessment of various cavitator shapes for high-speed supercavitating torpedoes: Geometry, flow-physics and drag considerations.* **Journal of Applied Fluid Mechanics**, **17(9)**, 2028-2044. IF: 1.3 (SCIE)
- 2024 Karn, A., Kar, A., Anand, R., Kanchi, R., Kataria, A., Singh, V. P. (2024). *The mustard bullets and the delightful culinary adventure in an Indian kitchen: High-Speed Visualizations, flow physics and mechanisms.* **Journal of Flow Visualization and Image Processing**, **31(2)**, 77-98. CiteScore: 1.3 (ESCI)
- 2024 Dhiman, P., Singh, V. P., Karn, A. (2024). *Experimental Analysis of Air Injection as a Mitigation Technique for Silt Erosion in Hydro Turbines.* **Renewable Energy and Sustainable Development**, **10(2)**, 345-365 (Scopus)
- 2023 Karn, A., Vyas, A., Upadhyay, A., Dwivedi, A. (2023). *Elegant computational frameworks for the analysis of cantilevers and beams.* **Journal of Engineering Education Transformations**, **37(1)**, 89-97. CiteScore: 1.8 (Scopus)
- 2023 Karn, A., Anand, R., Kataria, A., Singh, V. P. (2023). *Playing flute without lips? Tones of music lost in time - An investigation of the indigenous Bastar Flute.* **International Journal of Aeroacoustics**, **22**, 1-2. IF: 1.3 (SCIE)
- 2023 Karn, A., Mahawar, R., Dwivedi, P., Agarwal, R. (2023). *An Excel/VBA Teaching Learning Solution of Velocity Triangle Analysis of Hydraulic Turbines.* **Journal of Engineering Education Transformations**, **37(2)**, 40-50. CiteScore: 1.8 (Scopus)
- 2022 Singh, V. P., Jain, S., Karn, A., Kumar, A., Dwivedi, G., Meena, C. S., Cozzolino, R. (2022). *Mathematical Modelling of thermal, effective and exergetic efficiency evaluation of double pass parallel flow solar air heater.* **Sustainability** **2022**, **14(17)**, 10535. IF: 3.3 (SCI)

- 2022 Gaurav, K., Mittal, G., Karn, A. (2022). *On the Morphology of Elongated bubbles during their formation at submerged orifices.* **Chemical Engineering Science**, **250**, 117395. IF: 4.3 (SCI)
- 2022 Singh, V. P., Jain, S., Kumar, A., Dwivedi, G., Karn, A. (2022). *Heat transfer and friction factor correlations development for double pass solar air heater artificially roughened with perforated multi-V ribs.* **Case Studies in Thermal Engineering**, **39**, 102461. IF: 6.4 (SCI)
- 2022 Singh, V. P., Jain, S., Karn, A., Kumar, A., Dwivedi, G., Meena, C. S., Dutt, N., Ghosh, A. (2022). *Recent developments and advancements in solar air heater. A review.* **Sustainability** **2022**, **14**(19), **12149**. IF: 3.3 (SCI)
- 2022 Singh, V. P., Jain, S., Karn, A., Kumar, A., Dwivedi, G., Alam, T. (2022). *Experimental assessment of variation in open area ratio on thermohydraulic performance of parallel flow solar air heater.* **Arabian Journal for Science and Engineering**, **48**, 11695-11711. IF: 2.9 (SCI)
- 2022 Dwivedi, A., Sawant, G., Karn, A. (2022). *Computational solvers for iterative hydraulic loss calculations in pipe systems.* **Journal of Engineering Education Transformations**. CiteScore: 1.8 (Scopus)
- 2021 Kumar, A., Saha, A. K., Panigrahi, P. K., Karn, A. (2021). *The mechanism of vortex bifurcation vis-à-vis axial switching in rectangular synthetic jets.* **European Journal of Mechanics - B(Fluids)**, **86**, 78-89. IF: 2.5 (SCI)
- 2021 Bhat, A., Raghav, G., Karn, A. (2021). *Enhancement of Efficiency and Thermal Comfort in a Passenger Car Cabin Using Spanwise Inlet Vents.* **Engineering Journal**, **25**(2), 95-108. IF: 1.0 (ESCI)
- 2021 Yoon, K., Li, J., Shao, S., Karn, A., Hong, J. (2021). *Insights into the ventilation hysteresis shift due to flow unsteadiness in ventilated supercavitation.* **Experimental Thermal and Fluid Science**, **129**, 110472. IF: 3.3 (SCI)
- 2021 Karn, A., Ambs, J., Deo, S. S., Vyas, A., Dwivedi, A., Kumar, A. (2021). *On the enormous enhancement in water evaporation rates obtained by employing a different polar ambient.* **Heat Transfer**, **2021**, 1-12. IF: 2.6 (ESCI)
- 2020 Kumar, A., Saha, A. K., Panigrahi, P. K., Karn, A. (2020). *Implications of Velocity Ratio on the Characteristics of a Circular Synthetic Jet Flush Mounted on a Torpedo Model in Quiescent and Cross-flow Conditions.* **Journal of Applied Fluid Mechanics**, **13**(3), 1003-1013. IF: 1.3 (SCIE)
- 2020 Kumar, A., Saha, A. K., Panigrahi, P. K., Karn, A. (2020). *The effect of free surface upon a parallel submerged shallow synthetic jet.* **Fluid Dynamics Research**, **52**(4), 1-17. IF: 1.3 (SCIE)
- 2020 Karn, A., Singh Deo, S., Singh, R. K., Kumar, A. (2020). *A COMfortably Vented, Indigenously Designed (COVID) Fabric Helmet to Curb Infection Spread in Education, Healthcare and Other Community Settings.* **Health and Population: Perspectives and Issues**, **43**(4), **137–152**. (An invited 'Special Paper' published by NIHFWS, autonomous publication of Ministry of Health and Family Welfare, Government of India during COVID-19 pandemic)
- 2020 Karn, A., De, R., Kumar, A. (2020). *Some insights into drop impacts on a hydrophobic surface.* **Journal of Applied Fluid Mechanics**, **13**(2), 527-536. IF: 1.3 (SCIE)
- 2020 Chintala, V., Vikesh, S., Karn, A. (2020). *Efficiency and Effectiveness enhancement of an intercooler of two-stage air compressor by low-cost Al_2O_3 /water nanofluids.* **Heat Transfer**, **2020**, 1-18. IF: 2.6 (SCIE)
- 2020 Venkatramakrishnan, S. R., Pandey, J. K., Mondal, A. K., Karn, A. (2020). *Low Speed Wind Turbines for Power Generation: A review.* **Journal of Advanced Research in Fluid Mechanics and Thermal Sciences**, **67**(1), 146–169 (Scopus)
- 2019 Kumar, A., Saha, A. K., Panigrahi, P. K., Karn, A. (2019). *On the flow physics and vortex behavior of rectangular orifice synthetic jets.* **Experimental Thermal and Fluid Science**, **103**, 163-181. IF: 3.3 (SCI)

- 2019 Chintala, V., Sharma, A., Karn, A., Vardhan, H., Pandey, J. K. (2019). *Utilization of biomass-derived pyro-oils in compression ignition (CI) engines – Recent developments*. **Energy Sources, Part A: Recovery, Utilization and Environmental Effects**, **46(1)**, 986-1000. IF: 2.2 (SCI)
- 2019 Karn, A., Chintala, V., Kumar, S. (2019). *On the determination and variations in Sky temperature and its effect on solar cooker performance*. **Heat Transfer**, **48(5)**, 1830-1856. IF: 2.6 (ESCI)
- 2019 Karn, A., De, R., Kumar, A. (2019). *A Qualitative Inter-relation between Drop Impacts on a Solid Surface and the Interaction of a Droplet Placed Inside a Superheated Pool*. **Journal of Advanced Research in Fluid Mechanics and Thermal Sciences**, **60(1)**, 85-94. CiteScore: 2.8 (Scopus)
- 2018 Karn, A., Chawdhary, S. (2018). *On the synergistic interrelation between supercavity formation through vaporous and ventilated routes*. **International Journal of Multiphase Flow**, **104**, 1-8. IF: 3.8 (SCI)
- 2018 Kumar, A., Karn, A. (2018). *Qualitative and Quantitative characterization of a synthetic jet mounted on a convex torpedo-like surface under cross-flow*. **International Journal of Heat and Fluid Flow**, **74**, 198-208. IF: 3.1 (SCI)
- 2017 Kumar, S. S., Karn, A., Arndt, R. & Hong, J. (2017). *Internal flow measurements of a drop impacting a solid surface*. **Experiments in Fluids** **58:12**. IF: 2.5 (SCI)
- 2017 Cao, L., Karn, A., Arndt, R., Hong, J. (2017). *Numerical investigations into the pressure distribution within a ventilated supercavity*. **Journal of Fluids Engineering**, **139 (2)**, 021302. IF: 2.4 (SCIE)
- 2017 Karn, A., Rosiejka, B. (2017). *Air Entrainment Characteristics of Artificial Supercavities for Free and Constrained Closure Models*. **Experimental Thermal and Fluid Science**, **81**, 364-369. IF: 3.3 (SCI)
- 2017 Karn, A., Kumar, N., Aravindan, S. (2017). *Chemical Vapor Deposition Synthesis of Novel Indium Oxide Nanostructures in Strongly Reducing Growth Ambient*. **Journal of Nanostructures**, **7(1)**, 64-76. IF: 1.3 (ESCI)
- 2017 Shao, S., Karn, A., Ahn, B. K., Arndt, R., Hong, J. (2017). *A comparative Study of Supercavitation across two closed-wall water tunnel facilities*. **Experimental Thermal and Fluid Science**, **88**, 519-529. IF: 3.3 (SCI)
- 2016 Karn, A., Arndt, R., Hong, J. (2016). *An Experimental Investigation into Supercavity Closure Mechanisms*. **Journal of Fluid Mechanics**, **789**, 259-284. (Featured on the cover page of the Vol. 789 issue of JFM by Cambridge University Press). IF: 4.2 (SCI)
- 2016 Lee, S. J., Kawakami, E., Karn, A. & Arndt, R. (2016). *A Comparative study on the behaviors of ventilated supercavities between experimental models with different cavitator configurations*. **Fluid Dynamics Research**, **48**, 45506-45517. IF: 1.3 (SCIE)
- 2016 Karn, A., Shao, S., Arndt, R., Hong, J. (2016). *Bubble coalescence and breakup in turbulent bubbly wake of a ventilated hydrofoil*. **Experimental Thermal and Fluid Science**, **70**, 397-407. IF: 3.3 (SCI)
- 2016 Karn, A., Arndt, R., Hong, J. (2016). *Gas entrainment behaviors in the formation and collapse of a ventilated supercavity*. **Experimental Thermal and Fluid Science**, **79**, 294-300. IF: 3.3 (SCI)
- 2015 Karn, A., Monson, G., Ellis, C., Hong, J., Arndt, R., Gulliver, J. (2015). *Mass transfer studies across ventilated hydrofoils: A step towards hydroturbine aeration*. **International Journal of Heat and Mass Transfer**, **87**, 512-520. IF: 5.8 (SCI)
- 2015 Karn, A., Arndt, R., Hong, J. (2015). *Dependence of supercavity closure upon flow unsteadiness*. **Experimental Thermal and Fluid Science**, **68**, 493-498. IF: 3.3 (SCI)

- 2015 Karn, A., Ellis, C., Arndt, R., Hong, J. (2015). *An integrative image measurement technique for dense bubbly flows with a wide size distribution*. **Chemical Engineering Science**, **122**, 240-249. IF: 4.3 (SCI)
- 2015 Karn, A., Ellis, C., Milliren, C., Hong, J., Scott, D., Arndt, R., Gulliver, J. (2015). *Bubble size characteristics in ventilated hydrofoils with two aeration configurations*. **International Journal of Fluid Machinery and Systems**, **8(2)**, 73-83. IF: 0.66(Scopus)
- 2015 Karn, A., Ellis, C., Hong, J., Arndt, R. (2015). *Investigations into the turbulent bubbly wake of a ventilated hydrofoil: moving toward improved turbine aeration techniques*. **Experimental Thermal and Fluid Science**, **64**, 186-195. IF: 3.3 (SCI)
- 2012 Karn, A., Kumar, M., Singh, V. N., Mehta, B. R., Aravindan, S., Singh, J. P. (2012). *Growth of Indium Oxide and Zinc-doped Indium Oxide Nanostructures*. **Chemical Vapor Deposition**, **18**, 295-301. IF: 1.3 (SCI)

Selected Patents/Design Registrations/Copyrights

[Patents: 8 (Granted:1, Published:7); Designs: 8 (Granted:5; Filed:3); Copyrights: 9]

- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V.P. (2025). *Hybrid Cooking Burners*. (Filed) Design Registration # 461136-001. Design Office, Government of India.
- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V.P. (2025). *Hemispherical Shaped Hybrid Solar and Electric Cooking Burners*. Design Registration # 461138-001. Design Office, Government of India.
- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V.P. (2025). *Flat-Shaped Hybrid Solar and Electric Cooking Burners*. Design Application. (Filed).
- 2025 Rawat, A., Rana, A., Goswami, T., Karn, A., Singh, V.P. (2025). *Compact and Efficient Portable Air Heater with Integrated Regulator and Rechargeable Battery Power*. Indian Patent Application # 202411038059 A. (Published). Date of publication – 07/06/2024.
- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V.P. (2025). *Conical Shaped Hybrid Solar and Electric Cooking Burners*. Design Application. (Filed).
- 2025 Kumar, A., Karn, A., McGregor, C., Singh, V.P. (2025). *Smart Hybrid Solar Kitchen for Efficient Cooking Using Solar, Electric, and LPG Energy Sources*. Indian Patent. (Published).
- 2024 Karn, A., Kataria, A., Gupta, M., Basu, S., Anand, R. (2024). *Variable Vortex Smoke Ring Dispensing Device*. Patent # 550728. The Patent Office, Government of India. Date of Grant – Sep 23, 2024.
- 2024 Singh, P., Jain, H., Singh, P., Karn, A., Singh, V.P. (2024). *A foldable drone system*. Indian Patent Application # 202411043277 A. (Published). Date of publication – 05/07/2024.
- 2024 Singh, P., Jain, H., Singh, P., Karn, A., Singh, V.P. (2024). *A launching system for foldable drones*. Indian Patent Application # 202411047775 A. (Published). Date of publication – 05/07/2024.
- 2024 Singh, V.P., Rathore, G.S., Karn, A. (2023). *Solar Powered Drying System*. Indian Patent Application # 202411018464 A. (Published). Date of publication – 12/04/2024.
- 2024 Uniyal, V., Karn, A., Singh, V.P. (2024). *A universal testing setup for Archimedes screw turbine and a system thereof*. Indian Patent Application # 202411049384 A. (Published). Date of publication – 05/07/2024.
- 2024 Kaushik, A., Mishra, A., Gaurav, K., Karn, A. (2024). *A Passive Winged Atmospheric Descent Device with Auto-Rotation and Payload Capacity*. Indian Patent Application # 202411028678. (Published).
- 2022 Donga, R.K., Karn, A. (2022). *Bladeless kitchen exhaust fan*. Design Registration # 368060-001. Design Office, Government of India. (Granted).

- 2022 Karn, A., Karn, V., Kumar, A., Deo, Shashank S. (2022). *A Fabric Helmet Based Protective Gear*. Design Registration # 329398-001, Design Office, Government of India.
- 2022 Ganguly, S., Bhardwaj, D., Karn, A., Chakraborti, T. (2022). *Home Heating Apparatus*. Design Registration # 359641-001. Design Office, Government of India. (Granted).
- 2022 Karn, A., Kataria, A., Gupta, M., Basu, S., Anand, R. (2022). *(Vortex) Toy Gun*. Design Registration # 356385-001. Design Office, Government of India. (Granted).
- 2022 Karn, A., Badoni, P., Trivedi, T. (2022). *A Mentoring Scheme Including Personalized Notification of Course Performance to Students*. Copyright Registration # SW-15226/2022.
- 2021 Karn, A., Chandok, K., Joshi, M., Trivedi, T. (2021). *An Assignment Design and Transmission Portal to Inculcate Integrity and Self-Reliance*. Copyright Registration # SW-15072/2021.
- 2021 Thakur, S., Karn, A., Badoni, P. (2021). *ANURL: An Indigenous and Innovative URL Shortening Service*. Copyright Registration # SW-14748/2021.

Books

- 2025 Singh, V. P., Kumar, A., Karn, A. (2025). *Solar Thermal Fluids: Thermo-Fluid Dynamics in High-Temperature Applications*. **CRC Press, Taylor and Francis**.
- 2025 Karn, A., Singh, R. K. (2025). *Names That Count: The Cultural Calculus of Naming in India. How Indian Names Encode a Civilizational Legacy in Mathematics*. **Clever Fox Publishing**. ISBN - 9367073569

Selected Book Chapters/ Conference Proceedings

- 2026 Verma, M., Antia, V., Karn, A. (2026). *Computational Modeling and Aerodynamic Feasibility of Aircraft Concepts Inspired by Ancient Indian texts Using Biomimetic Design Principles*. **Lecture Notes in Mechanical Engineering**. Proceedings of the 1st International and 4th National Space Convention, UPES Dehradun, Uttarakhand. Springer, Singapore.
- 2024 Dhiman, P., Bhat, A., Karn, A. (2024). *The Efficacy of Air Injection in Mitigating Silt Erosion on Hydroturbine Blades: A Computational Study*. In: Hodge, B.M., Prajapati, S.K. (eds) **Proceedings from the International Conference on Hydro and Renewable Energy**. ICHRE 2022. Lecture Notes in Civil Engineering, vol 391. (Chapter 49). Springer, Singapore, pp. 437-444.
- 2024 Sanjeevaiah, P.K., Donga, R.K., Karn, A. (2024). *Performance Evaluation of an Undershot Water Wheel Turbine Using CFD*. In: Hodge, B.M., Prajapati, S.K. (eds) **Proceedings from the International Conference on Hydro and Renewable Energy**. ICHRE 2022. Lecture Notes in Civil Engineering, vol 391. (Chapter 39), pp. 343-350.
- 2023 Verma, M., Karn, A. (2023). *Thermal Energy Applications in Net Zero Energy Buildings*. In: **Thermal Energy Systems: Design, Computational Techniques and Applications**, CRC Press.
- 2023 Rajkumar, R., Gaurav, K., Karn, A., Kumar, V., Shukla, H. (2023). *Numerical Investigation of the Effect of Liquid Temperature on Supercavitation*. In: Narendranath, S., Mukunda, P.G., Saha, U.K. (eds) **Recent Advances in Mechanical Engineering**. Lecture Notes in Mechanical Engineering. Springer, Singapore. Chapter 02, pp. 19-27.
- 2023 Agarwal, R., Dwivedi, P., Mahawar, R., Karn, A. (2023). *Design and Analysis of Longitudinal Butt Joints Using an Excel/VBA Computational Tool*. In: Narendranath, S., Mukunda, P.G., Saha, U.K. (eds) **Recent Advances in Mechanical Engineering**. Lecture Notes in Mechanical Engineering. Springer, Singapore. Chapter 12, pp. 119-130.

- 2023 Dwivedi, A., Sawant, G., Vyas, A., Karn, A. (2023). *Effective Computational Tools for Teaching and Learning of Heat Transfer Through Extended Surfaces*. In: Narendranath, S., Mukunda, P.G., Saha, U.K. (eds) **Recent Advances in Mechanical Engineering**. Lecture Notes in Mechanical Engineering. Springer, Singapore. Chapter 13, pp. 131-140.
- 2018 Karn, A., Narula, V., Arndt, R.E.A., Hong, J. (2018). *Insights into Ventilation Demand Estimation for High-Speed Supercavitating Underwater Vehicles*. In: Singh, M., Kushvah, B., Seth, G., Prakash, J. (eds) **Applications of Fluid Dynamics**. Lecture Notes in Mechanical Engineering. Springer, Singapore. Chapter 10, pp. 683-690.
- 2016 Karn, A., Huang, R., Shao, S., Arndt, R., Hong, J. (2016). *Probing into physics of ventilation demand for supercavitating underwater vehicles*. **Proceedings of 31st Symposium on Naval Hydrodynamics**, Monterey, CA.
- 2016 Ahn, B.K., Jeong, S.W., Kim, J.H., Shao, S., Karn, A., Hong, J., Arndt, R. (2016). *An experimental Investigation of artificial supercavitation*. **Proceedings of 31st Symposium on Naval Hydrodynamics**, Monterey, CA.
- 2016 Karn, A., Gulliver, J. S., Monson, G., Ellis, C., Arndt, R., Hong, J. (2016). *Gas transfer in a bubbly wake flow*. **IOP Conf. Ser.: Earth Environ. Sci.**, **35(1)**, 012020. Proceedings of the 7th International Symposium on Gas Transfer at Water Surfaces, May 18–21, 2015, Seattle, Washington, USA.
- 2015 Karn, A., De, R., Arndt, R., Hong, J. (2015). *Closure mechanisms of ventilated supercavities under steady and unsteady flows*. **Journal of Physics: Conference Series**, **656(1)**, 012145. Proceedings of the 9th International Symposium on Cavitation (CAV2015).
- 2014 Ellis, C., Karn, A., Hong, J., Lee, S., Kawakami, E., Scott, D., Gulliver, J., Arndt, R. (2014). *Measurements in the wake of a ventilated hydrofoil: a step towards improved turbine aeration techniques*. **IOP Conf. Ser.: Earth Environ. Sci.**, **22**, 062009. Proceedings of the 27th IAHR Symposium on Hydraulic Machinery and Systems (IAHR 2014).

Invited Talks/ Workshops/ FDPs delivered

- 2025 *Role of Ayurveda in Indian Knowledge Systems: A holistic perspective*. **Invited talk on World Science Day by Ozone Student Society**, UPES Dehradun, Nov. 13, 2025
- 2025 *Wheels, Wings, and Waves: Interlink between Automotive, Marine and Aeronautical Industry*. **Workshop on Advances in Automotive Industry (WAAI'25; with GarageX)**, UPES Dehradun, Sep. 15, 2025
- 2025 *Hybrid Propulsion System*. **ASME Gas Turbine India Group — Student Seminar Series**, University of Petroleum and Energy Studies (UPES), Dehradun, Sep. 26, 2025. Event page
- 2025 *Survey of Indian Knowledge System: Emulation of Natural Systems for Science, Technology and Holistic Education*. **Faculty Development Program**, Ramanujan College, University of Delhi (under UGC–Malaviya Mission Teacher Training Program, MMTTP), Aug. 29, 2025.
- 2025 *Nakshatra – A Journey Through Ancient Astronomy and Heritage*. **Guest Speaker**, UPES Dehradun (organized by Infinity Space Club, Dharohar & ASI Dehradun Circle), Apr. 26, 2025.
- 2025 *Mathematical Consciousness in India: Numbers and Beyond – A Socio-Mathematical Exploration through Names*. **Ganit Sammelan**, IIT Gandhinagar, 19-22 Jan, 2025
- 2024 *Identification of Indian Knowledge Systems in Literature for Inclusion in Science and Mathematics Textbooks*. Two-day workshop by **Department of Education in Science and Mathematics, NCERT**, New Delhi, 03-04 Sep. 2024.
- 2024 *From Bio-inspiration to Biomimicry: Exploring the Indian roots and Implications for Modern Education*. Two-day workshop at **Inter-University Accelerator Centre, New Delhi**, 11-12 July 2024.

- 2023 *The Indian Tradition of Computation and Algorithms*. **Prodigy 2023: Sanskriti, UPES ACM-W Student Chapter**, Dehradun, 25th April 2023.
- 2020 *Unravelling the flow physics of transport – The interlink between automotive and other modes*. National Faculty Development Program on “Technological Evolution and Implementation of Modern Tools in Automotive Engineering”, 30th June, 2020.
- 2019 *Research Perspectives on Extracting Kinetic Hydro Power*. **ONGC Energy Center, New Delhi, India**; Feb 6, 2019.
- 2019 *Case Studies on Hydropower Research: American and Indian Perspectives*. **TATA Power Hydros, Khopoli, Maharashtra, India**; Jan 2, 2019.
- 2018 *Multiphase Flows in Turbomachinery*. Jaipur Institute of Engineering and Technology, Jaipur, Rajasthan, India, April 2018.
- 2016 *Applications of Multiphase Flows*. School of Mechanical, Materials and Energy Engineering, **Indian Institute of Technology Ropar, Punjab, India**, May 3, 2016.
- 2016 *Bubbly Flow Physics for Applications in Aerated Hydroturbines and Underwater Transport*. Department of Fire Protection Engineering, **Worcester Polytechnic Institute, Worcester, Massachusetts, USA**, April 28, 2016.
- 2016 *Multiphase Flows in Fluid Machinery*. Department of Mechanical Engineering, **California State University Fullerton, Los Angeles, California, USA**, April 5, 2016.
- 2016 *Computational Simulations in Mechanical Engineering*. Engineering and Technology Department, **University of Wisconsin Stout, Menomonie, Wisconsin, USA**, March 29, 2016.
- 2015 *A Robust Image Analysis Approach for High Void Fraction Gas-Liquid Flows*. 3rd World Congress on Petrochemistry and Chemical Engineering, Atlanta, USA, Nov. 30, 2015.

Selected Conference Presentations

Indian Knowledge Systems

- 2025 Karn, A., Kumar, V., Saswadkar, M. A., Jindal, A., Sharma, S., Tahilramani, A., & Sai, A. T. (2025). *Flight Dynamics of Arrows in Ancient Indian Text*. **1st International Space Convention and 4th National Space Convention**, School of Advanced Engineering, UPES Dehradun, 13–15 November 2025.
- 2023 Karn, A., Pandey, M., Badoni, P. (2023). *From Bio-inspiration to Biomimetics in Indian Knowledge Systems: Inspirations for a Sustainable Future*. Indian Knowledge Systems: Issues, Challenges and Way Forward.
- 2023 Karn, A., Pandey, M., Badoni, P. (2023). *IKS through gamification and game design: A pedagogical approach for integration in modern mathematics education*. Indian Knowledge Systems: Issues, Challenges and Way Forward.
- 2023 Karn, A., Pandey, M., Badoni, P. (2023). *Legacies unnoticed in the modern education system: The curious case of the mathematical names of Indians and the ways ahead*. Indian Knowledge Systems: Issues, Challenges and Way Forward.
- 2023 Karn, A. (2023). *The geometrical names of Indians: An imprint of a legacy*. International Conference on Indian Knowledge Systems, Varanasi, India, March 2023.
- 2023 Pandey, M., Karn, A. (2023). *Understanding of solar phenomena in the pre-telescopic era: An Indian perspective*. National Youth Conference on Indian Knowledge Systems, Thiruvananthapuram, Kerala, India.
- 2023 Verma, M., Antia, V., Karn, A. (2023). *An investigation of the bio-inspired and biomimetic design principles of the Bṛhad Vaimānika Śāstra*. National Youth Conference on Indian Knowledge Systems, Thiruvananthapuram, Kerala, India.

Engineering and Applied Sciences

- 2025 Sanjeevaiah, P. K., Doonga, R. K., Karn, A. (2025). *Performance Enhancement of an Undershot Water Wheel: Experimental Insights into Blade Design and Count Optimization in a Lab-Scale Study*. **International Conference on Advances in Sustainable Solutions for Energy Transitions**, IIT Guwahati, 2-4th Jan 2025. **(First Prize in Oral Presentation Category)**.
- 2025 Karn, A., Kumar, V., Nagi, D., & Kaur, A. (2025). *Numerical Investigation of Optimal Parameters for Airflow Control over an Airfoil with Vortex Generators and Self-Suction Mechanism*. **1st International Space Convention and 4th National Space Convention**, School of Advanced Engineering, UPES Dehradun, 13–15 November 2025.
- 2025 Uniyal, V., Singh, V. P., Karn, A. (2025). *Effects of Flow Rate, Rotational Speed and Load on the Power Output of Archimedes Screw Turbines for Ultra-Low Head Remote Power Generation*. **International Conference on Advances in Sustainable Solutions for Energy Transitions**, IIT Guwahati, 2-4th Jan 2025.
- 2025 Karn, A., Kumar, V., Saswadkar, M. A., Jindal, A., Sharma, S., Tahilramani, A., & Sai, A. T. (2025). *Flight Dynamics of Arrows in Ancient Indian Text*. **1st International Space Convention and 4th National Space Convention**, School of Advanced Engineering, UPES Dehradun, 13–15 November 2025.
- 2024 Srivastava et al. (2024). *Control of Wing Rock Oscillations of a Delta Wing through Geometric Modification*. **Symposium on Applied Aerodynamics and Design of Aerospace Vehicles**, Thiruvananthapuram.
- 2023 Bhat, A., Dhiman, P., Karn, A. (2023). *The efficacy of air injection in mitigating silt erosion on hydro turbine blades: a computational study*. **International Conference on Hydro and Renewable Energy (ICHRE)**, IIT Roorkee, October 17-18.
- 2023 Dhiman, P., Vyas, A., Singh, V. P., Karn, A. (2023). *Hydroturbine Blade and Guide Vane Erosion due to Silt: Investigating Size Distribution and Constituent Analysis*. **2nd International Conference on Aspects of Mechanical and Materials Engineering**, India.
- 2023 Karn, A., Anand, R. (2023). *Experimental Investigations of the Dynamics of Free Rotation using Smartphones for At-Home Science/Engineering Labs*. **2nd GUS Academic Summit 2023** (Transformation for Future-proof Education), Dehradun, India.
- 2022 Kumar, P., Donga, R., Karn, A. (2022). *Performance evaluation of an undershot waterwheel turbine using CFD*. **International Conference on Hydro and Renewable Energy (ICHRE)**, IIT Roorkee, October 17-18.
- 2020 Vyas, A., Singh, D., Karn, A. (2020). *Secondary fluid injection in turbomachines*. **Advances in Chemical Engineering**, Dehradun, India.
- 2020 Mahawar, R., Singh, P., Karn, A. (2020). *Novel surface aerators for water quality enhancement in hydropower impoundments*. **Tech Symposium**, UPES Dehradun. **(Best Paper Award)**.
- 2020 Kar, A., Dhebar, A., Khan, K., Karn, A. (2020). *Visualization of bubble generation upon interaction with ultrasonic horn in liquids at different carbonation levels*. **Advances in Chemical Engineering**, Dehradun, India.
- 2020 Mahawar, R., Dwivedi, P., Vyas, A., Kumar, A., Karn, A. (2020). *Rewriting the cavitation story in engineering*. **Advances in Chemical Engineering**, Dehradun, India.
- 2019 Mittal, G., Donga, R. K., Kumar, A., Karn, A. (2019). *Identification of silt erosion areas on Francis turbine guide vanes through numerical simulation*. **46th National Conference on Fluid Mechanics and Fluid Power**, PSG Institute of Technology, Coimbatore, India.
- 2019 Donga, R. K., Kumar, A., Karn, A. (2019). *A numerical study on the bifurcation of rectangular orifice synthetic jets*. **46th National Conference on Fluid Mechanics and Fluid Power**, PSG Institute of Technology, Coimbatore, India.
- 2018 Srivastava, A., Mishra, A., De, R., Chintala, V., Kumar, A., Karn, A. (2018). *Some insights into drop impacts on a hydrophobic surface*. **7th International and 45th National Conference on Fluid Mechanics and Fluid Power**, IIT Bombay, India.

- 2018 Chintala, V., Kumar, S., Kumar, P., Kumar, A., Karn, A. (2018). *Development of an after-treatment device with low-cost catalysts for NO_x emission reduction from diesel engines.* **7th International and 45th National Conference on Fluid Mechanics and Fluid Power**, IIT Bombay, India.
- 2018 Karn, A., Ragunath, S., De, R., Chintala, V., Kumar, A. (2018). *High-speed imaging reveals a close resemblance between drop impacts on a solid surface and the interaction of a cold droplet placed upon a hot liquid pool.* **7th International and 45th National Conference on Fluid Mechanics and Fluid Power**, IIT Bombay, India.
- 2017 Karn, A. (2017). *Exploring the relationship between natural and artificial modes of supercavitation.* **44th National Conference on Fluid Mechanics and Fluid Power**, Amritapuri, Kollam, Kerala, India.
- 2016 Karn, A., Narula, V., Arndt, R., Hong, J. (2016). *Insight into Ventilation Demand Estimation for High-speed Supercavitating Underwater Vehicles.* **3rd International Conference on Applications of Fluid Dynamics**, Dhanbad, India.
- 2016 Karn, A., Rosiejka, B., Narula, V., Chaturvedi, A. (2016). *Some Insights into Air Entrainment Characteristics of Artificial Supercavities.* **6th International and 43rd National Conference on Fluid Mechanics and Fluid Power**, Allahabad, India.
- 2016 Karn, A., Shao, S., Arndt, R. E. A., Hong, J. (2016). *Probing into physics of ventilated demand for supercavitating underwater vehicles.* **31st ONR Symposium on Naval Hydrodynamics**, Monterey, CA.
- 2016 Ahn, B.K., Jeong, S.W., Kim, J.H., Shao, S., Karn, A., Hong, J., Arndt, R. (2016). *Experimental Investigation of artificial supercavitation.* **31st ONR Symposium on Naval Hydrodynamics**, Monterey, CA.
- 2015 Karn, A., Monson, G. M., Ellis, C., Hong, J., Arndt, R. E. A., Gulliver, J. (2015). *Oxygen transfer across ventilated hydrofoils.* **7th International Symposium on Gas Transfer at Water Surfaces**, Seattle, USA.
- 2015 Karn, A., Arndt, R. E. A., Hong, J. (2015). *Closure mechanisms of ventilated supercavities under steady and unsteady flow conditions.* **9th International Symposium on Cavitation**, Switzerland.
- 2015 Karn, A., Arndt, R. E. A., Hong, J. (2015). *On the relationship between supercavity closures, air entrainment and internal flows.* **68th Annual Meeting of APS-DFD**, Boston.
- 2015 De, R., Karn, A., Noonan, J., Rosiejka, B., Arndt, R., Hong, J. (2015). *Characterization of drop impact based on internal flow quantification.* **68th Annual Meeting of APS-DFD**, Boston, USA.
- 2015 Karn, A., Arndt, R. E. A., Hong, J. (2015). *A robust image analysis approach for high void fraction gas-liquid flows.* **3rd World Congress on Petrochemistry and Chemical Engineering**, Atlanta, USA.
- 2015 Scott, D., Sabourin, M., Beaulieu, S., Ellis, C., Karn, A. (2015). *Optimizing dissolved oxygen transfer in auto-venting turbines.* **HydroVision International 2015**, Portland, USA.
- 2015 Zhao J., Han H. S., Karn, A., McMurtry P. (2015). *Development of an Ultrafine Condensation Particle Counter (UCPC) for Measurements of Particles Down to 1 nm in Diameter.* **34th American Association for Aerosol Research Conference**, Minneapolis, Minnesota.
- 2014 Karn, A., Ellis, C., Arndt, R. E. A., Hong, J. (2014). *Bubble size measurements in a bubbly wake.* **67th Annual Meeting of APS-DFD**, San Francisco, USA.
- 2014 Arndt, R. E. A., Karn, A., Hong, J. (2014). *A unified physical model to explain supercavity closure.* **67th Annual Meeting of APS-DFD**, San Francisco, USA.

- 2014 Ellis, C., Karn, A., Hong, J., Lee, S. J., Kawakami, E., Scott, D., Gulliver, J., Arndt, R. E. A. (2014). *Measurements in the wake of a ventilated hydrofoil: A step towards improved turbine aeration techniques*. **27th IAHR Symposium on Hydraulic Machinery and Systems**, Montreal, Canada.
- 2014 Lee, S. J., Kawakami, E., Karn, A., Arndt, R. E. A. (2014). *Experimental investigation on the behaviors of ventilated supercavities*. **Proceedings of the annual autumn meeting of the SNAK (The Society of Naval Architects of Korea)**, Changwon, Korea, 2014.
- 2013 Kawakami, E., Lee, S. J., Karn, A., Hong, J., Arndt, R. E. A. (2013). *Improved performance with ventilation*. **66th Annual Meeting of APS-DFD**, Pittsburgh, USA.
- 2013 Arndt, R. E. A., Lee, S. J., Kawakami, E., Hong, J., Karn, A. (2013). *Ventilation of a hydrofoil wake*. **66th Annual Meeting of APS-DFD**, Pittsburgh, USA.
- 2011 Karn, A., Aravindan, S., Singh, V. N. (2011). *Metallographic analysis of Polyhedral Indium Zinc Oxide Mesocontainers*. **International Metallographic Contest 2011**, American Society of Metals, Aug. 7 – 11, 2011, Nashville, TN, USA.
- 2011 Karn, A., Aravindan, S., Singh, V. N., Mehta, B. R. (2011). *Synthesis, shape evolution and growth mechanism of Polyhedral Indium Zinc Oxide Mesocontainers*. **International Metallographic Contest 2011**, American Society of Metals, Aug. 7 – 11, 2011, Nashville, TN, USA.

Doctoral Supervision

- 2026 **Amit Kumar** (Co-Guide)
Thesis: Design, Development and Performance Analysis of Heaters for Hybrid Solar Kitchen Applications.
 PhD in Mechanical Engineering — Awarded, 2026
- 2026 **Vipin Uniyal** (Primary Guide)
Thesis: Design, Development and Performance Analysis of Archimedes Screw Turbine.
 PhD in Mechanical Engineering — Awarded, 2026
- 2025 **Prashant Dhiman** (Sole Guide)
Thesis: Investigations on Performance of Air-Injection Technique in Mitigating Silt Erosion in Hydro-Turbines.
 PhD in Mechanical Engineering — Awarded, 2025
- 2025 **Prashant Kumar Sanjeeviah** (Primary Guide)
Thesis: Moving Towards Environment-Friendly Hydropower: Development and Performance Analysis of Hydrokinetic Water-Lifting Technology for Hilly Areas.
 PhD in Mechanical Engineering — Awarded, 2025
- 2024 **Kumar Gaurav** (Sole Guide)
Thesis: Systematic Investigations on Strategies for Enhancement of Supercavity Size.
 PhD in Aerospace Engineering — Awarded, 2024

Work with Industry

- 2025 **Narendra Explosives**, Dehradun, Uttarakhand, India
 Instrumental in establishing a Mutual Non-Disclosure Agreement (MoU) between UPES and Narendra Explosives Ltd. for collaborative research and development in ammunition and explosive technologies.

- 2024 **Ecosense Appliances Pvt. Ltd.**, Aurangabad, Maharashtra, India
Initiated and co-led the establishment of the Clean Combustion and Energy Systems Center at UPES in collaboration with Ecosense Appliances Pvt. Ltd., facilitating an industry-academia partnership for experimental research in biomass stoves, incinerators, and sustainable combustion technologies.
- 2018–2019 **TATA Power Hydros**, Khopoli, Bhivpuri and Bhira, Maharashtra, India
Worked on problems related to water quality.
- 2013–2015 **ALSTOM Renewable Power Canada Inc.**, Sorel-Tracy, Quebec, Canada
Suggested design and operational strategies for ventilated hydro turbine blades and investigated bubble size and mass transfer characteristics under different hydrodynamic conditions through water-tunnel experiments.
- 2011–2013 **TSI Inc.**, Shoreview, Minnesota, USA
Conducted extensive bench-top laboratory experiments, field measurements, and data analysis for the design and development of a novel particle counter capable of measuring particles as small as 1 nm.

Professional Life Memberships

- National Society of Fluid Mechanics and Fluid Power, IIT Bombay, India (Membership # LM652).
- Indian Society for Heat and Mass Transfer, IIT Madras (Membership # 1980).
- The Institution of Engineers (India) (Membership # M-1829955).
- The Aeronautical Society of India (Membership # M-21214).
- Astronautical Society of India, U.R. Rao Satellite Centre, ISRO (Membership # LM3638).
- Indian Society for Technical Education (Membership # LM143418).
- Vijnana Bharati, India (Member since Dec. 18, 2023).
- Society for History of Science, Kolkata (Membership # LM55).
- Indian Society for History of Mathematics, University of Delhi.

Professional Development Training / Workshops / Courses Attended

- Two-day workshop on teaching innovations: *Elevate Workshop*, Centre for Teaching and Learning, UPES Dehradun, Feb. 2026.
- Two-day workshop on *Funding Opportunities and Project Writing*, Centre for Interdisciplinary Research and Innovation, UPES, Apr. 2022.
- Second Online International Workshop on In-stream Hydrokinetic Device, AHEC, IIT Roorkee, Oct. 2021.
- Digital Transformation, Coursera course by the University of Virginia and Boston Consulting Group, Jul. 2020.
- Virtual Teacher Specialization, five-course Coursera specialization with Capstone Project, UC Irvine, 2020.
- Performance Assessment in the Virtual Classroom, Coursera, UC Irvine, Jun. 2020.
- Advanced Instructional Strategies in the Virtual Classroom, Coursera, UC Irvine, Jun. 2020.
- Emerging Trends and Technologies in the Virtual K-12 Classroom, Coursera, UC Irvine, Jun. 2020.
- Foundations of Virtual Instruction, Coursera, UC Irvine, May 2020.
- Faculty Development Program on Outcome-Based Education, UPES, May 25-27, 2020.
- Programming for Everybody (Getting Started with Python), Coursera, University of Michigan, Apr. 2020.
- One-day Workshop on R, IIT Bombay under NMEICT, Nov. 9, 2019.
- Practicum for Future Faculty: Preparing Future Faculty, Center for Educational Innovation & Graduate School, University of Minnesota Twin Cities, Jan.-May 2016.
- Teaching in Higher Education: Preparing Future Faculty, Center for Educational Innovation & Graduate School, University of Minnesota Twin Cities, Aug.-Dec. 2015.